

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA1603: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

C02: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

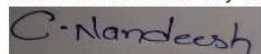
Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q 1						
Q 2						
Q 3						
Q 4						
Q 5						

Code of Conduct:

I **C.Nandeesh(1925250150)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

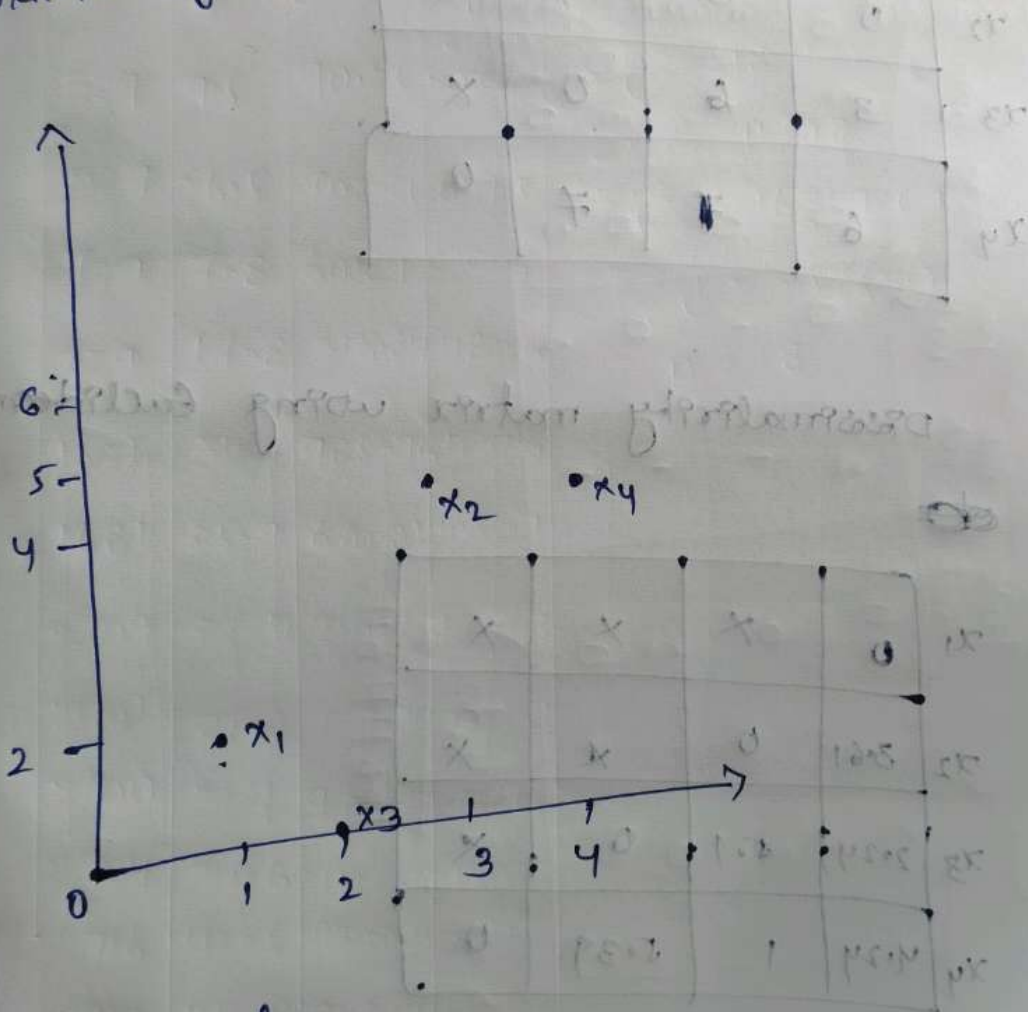
RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Activity-5

Minkowski Distance

consider the following point $(1, 2)$ $(3, 5)$ $(2, 0)$ $(4, 5)$ construct the data matrix & similarity matrix by Manhattan & Supremum measures



data matrix:-

point	Attr-1	Attr-2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

Dissimilarity matrix using Manhattan method.

	x_1	x_2	x_3	x_4
x_1	0	7	4	4
x_2	5	0	1	4
x_3	3	6	0	4
x_4	6	7	7	0

Dissimilarity matrix using Euclidean

~~the~~

	x_1	x_2	x_3	x_4
x_1	0	7	4	4
x_2	3.61	0	1	4
x_3	2.24	5.1	0	4
x_4	4.24	1	5.39	0

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$d = \sqrt{(2-1)^2 + (5-2)^2}$$

~~1.5.19~~

Distance matrix using supremum.

	x_1	x_2	x_3	x_4
x_1	0	x	x	x
x_2	3	0		x
x_3	2	5	0	x
x_4	3	1	5	0

(3,5) (1,2)

$$d = \max |x_1 - x_2|, |5 - 2|$$

$$= 3$$

(2,0) (1,2)

using cosine similarity.

Document	team coach	hockey	base ball	do ccs	pe nalty	score	win	document
1	5	0	0	2	0	0	2	0
2	3	0	0	1	0	0	1	0
3	0	7	0	1	0	0	3	0
4	0	1	2	1	2	2	0	0

$$\cos(d_1, d_2) = \frac{(d_1 \cdot d_2)}{\|d_1\| \|d_2\|}$$

Ex. 1 find the similarity b/w d_1 & d_2

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$d_1 \cdot d_2 = 25$$

$$\begin{aligned} \|d_1\| &= 5 \times 5 + 3 \times 3 + 2 \times 2 + 2 \times 2 \\ &= 25 + 9 + 4 + 4 \\ &= \sqrt{42} \\ &= 6.481 \end{aligned}$$

$$\begin{aligned} \|d_2\| &= 3 \times 3 + 2 \times 2 + 1 + 1 + 1 + 1 \\ &= 9 + 4 + 4 \\ &= \sqrt{17} \\ &= 4.12 \end{aligned}$$

$$= \frac{25}{(6.481)(4.12)}$$

$$\boxed{= 0.9356}$$